Lctr 5: Basic System Properties

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Focus: Section 1.6 (Pages 44 to 55)

5.1 Memory vs. Memoryless Systems

5.1.1 Memoryless systems

The output at any time depends only on the input at that same time.

Example: The ReLU activator of neural network:

$$y[n] = \begin{cases} x[n], & x[n] \geq 0 \\ 0, & \text{otherwise} \end{cases}.$$

5.1.2 System with memory

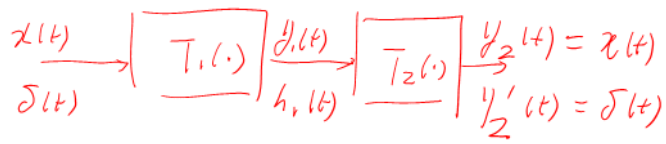
The output depends on past or future values of the input.

Example:

- $y_1[n] = x[n] + \alpha y_1[n-1]$: Current output depends on previous output in addition to the current input.
- $y_2[n] = x[n] - \beta x[n-1]$: Current output depends on previous input in addition to the current input.

5.2 Invertibility and Inverse Systems

5.2.1 Definitions of inverse systems



Two similar definitions:

- A system is **invertible** if distinct inputs lead to distinct outputs.
- A system $T_1(\cdot)$ is invertible if there exist another system $T_2(\cdot)$ such that the impulse response of the cascade of $T_1(\cdot)$ and $T_2(\cdot)$ is unit impulse.
  - For this case, $T_1(\cdot)$ and $T_2(\cdot)$ are the inverse system of each other.

See Fig. 1.45 on p. 46 for examples of inverse systems.

In many real applications, we cannot find an inverse system of a system; yet, we can find a system such that the cascade of the original system and the new system forms an all-pass filter. This can be used in designing mini-phase filters. This will be discussed in CEC 410.

5.2.2 Finding an inverse system

We will discuss z-transforms in the near future. With z-transform, System 1: $y_1[n] = x[n] + \alpha y_1[n-1]$ will have a system function

$y_1[n] - \alpha y_1[n-1] = x(n)$

$Y_1(z) - \alpha z^{-1} Y_1(z) = X(z)$

$H_1(z) = \frac{Y_1(z)}{X(z)}$

$$H_1(z) = \frac{1}{1 - \alpha z^{-1}}.$$

Similarly, System 2: $y_2[n] = x[n] - \beta x[n-1]$ will have a system function

$$H_2(z) = 1 + \beta z^{-1}.$$

If we choose $\alpha = -\beta$, we will have $H_1(z) \cdot H_2(z) = 1$. As we will learn later; the z-transform of the DT unit impulse is 1, which means these two systems will be the inverse system of each other.

5.3 Causality

5.3.1 Definition of a causal system

Two similar definitions:

- A system is **causal** if the output at any time depends only on present and past values of the input.
- A system is **causal** if the output at any time does not depend on future input.

An easy way to see if a system is causal is to see its impulse response:

- The impulse response is the response of the system to a unit impulse, which does not have non-zero value for $n < 0$.
- If the impulse response of a system has non-zero value for $n < 0$, it will not be a causal system—the non-zero value at $n < 0$ is caused by input at $n = 0$, which means that the output value depends on future values.

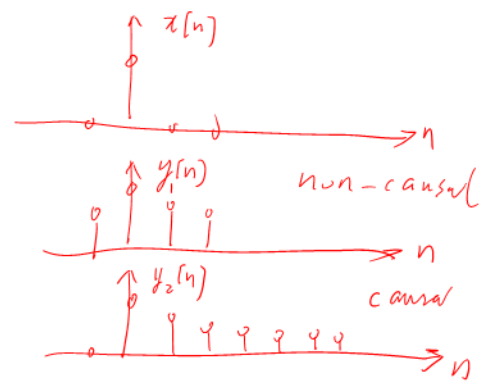
~~A property of easy way to see if a system~~ ### Examples of causal/noncausal systems

- $y_1[n] = x[n] + \alpha y_1[n-1]$:

  Causal if $y[n] = 0$ for $n < 0$
- $y_2[n] = x[n] - \beta x[n-1]$: Causal
- $y_3[n] = x[n] - \beta x[n+1]$: Non causal
- The ReLU activator:

$$y[n] = \begin{cases} x[n], & x[n] \geq 0 \\ 0, & \text{otherwise} \end{cases}.$$

Causal



5.4 Stability

5.4.1 Definition of a stable system

Informally, a **stable system** is one where small inputs do not lead to diverging responses.

5.4.1.1 BIBO stable

Formal Definition (BIBO): If the input is bounded, the output must also be bounded.

5.4.2 Examples of stable/non-stable systems

Example: A pendulum is stable; an inverted pendulum is unstable.

5.5 Time Invariance

5.5.1 Definitions of time invariance

A system is **time-invariant** if its characteristics do not change over time.

5.5.2 Verifying a system is time invariant or not

A time shift in the input signal must result in an identical time shift in the output signal.

5.6 Linearity

5.6.1 Definition of linearity

A system is **linear** if it possesses the property of **superposition**.

5.6.2 Additivity

Response to $x_1(t) + x_2(t)$ is $y_1(t) + y_2(t)$.

5.6.3 Scaling

Homogeneity: Response to $ax_1(t)$ is $ay_1(t)$.

5.6.4 Examples of linear/nonlinear systems

For a linear system, a zero input must result in a zero output.

Here we use the ASR system as an example:

- **Memory**:
  - Mel-freq vector calculation is memoryless.
  - Phoneme estimation has memory.
- **Causality**

  - A streaming ASR system is causal.
  - A batch-processing ASR system is non-causal.

- **Stability**
  - The system is stable.

- **Linearity**
  - The entire system is non-linear.
  - Components of the system can be linear. For example, the FFT part.

5.7 Exercises

Exercise 1: Consider the following systems

$$① \quad T_1(x[n]) = \frac{1}{x[n+1]}.$$

$$② \quad T_2(x[n]) = x[n] + 0.5x[n+1].$$

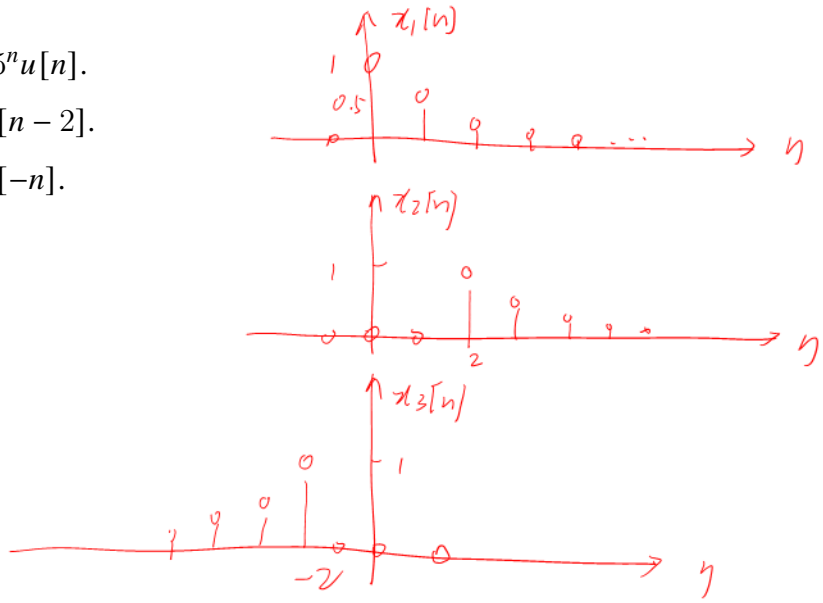
Determine if each system is

- Invertible. ① Yes, since different input values lead to different output values ② Yes, see Page 2.
- Causal.
- BIBO stable. ① No, since the output at n depends on input at n+1, future value. ② No, same as above
- Time invariant.
- Linear.
- Memoryless. ① No, since $\lim_{x[n+1]\to\infty} \frac{1}{x[n+1]} = \infty$ ② Yes.

Both yes

① No due to inverse relation ② Yes.

Both no.

Exercise 2 (preparation for convolution): Sketch the waveform of the following signals:

- $x_1[n] = 0.5^n u[n]$.
- $x_2[n] = x_1[n-2]$.
- $x_3[n] = x_2[-n]$.